

FUNDACIÓN FULGOR

CMOS Course

Homework N°2

Professors: Michel Saenz.
Santiago Castro.

Student: Monja, Ernesto Joaquín (43.873.728)

Year 2026

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1 Introduction

This report presents the small-signal alternating current (AC) analysis for the assigned *CMOS* amplifier circuit shown in Fig. 1.

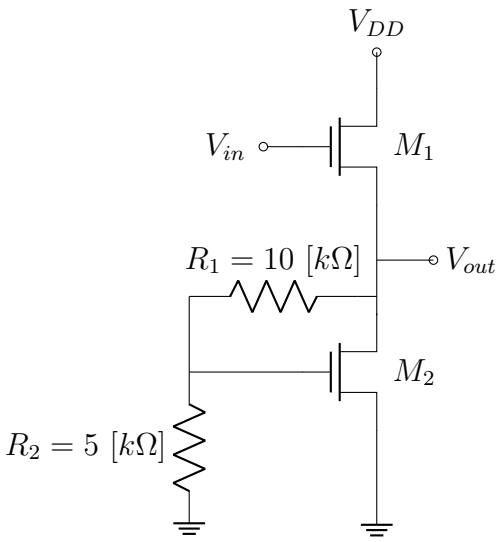


Figure 1: Circuit proposed to analyze.

The circuit topology in Fig. 1 is identified as a *Common-Drain* amplifier, also known as a *Source Follower* [1, p. 331]. In this configuration, the input signal is applied to the gate of transistor M_1, and the output is taken at its source. The transistor M_2 acts as an active load; however, unlike a standard source follower with a fixed bias, its gate is tied to a local negative feedback loop formed by the voltage divider R_1 and R_2. The objective of this report is to derive the complete small-signal voltage gain A_v considering channel-length modulation effects (this is assuming that $r_o = 5\ [k\Omega]$ for both transisors) and lastly we´ll use the following trans-conductance values for M_1 and M_2:

- $g_{m1} = 1\ [mS]$
- $g_{m2} = 3\ [mS]$

2 Development and Small-Signal Analysis

To accurately model the circuit in the small-signal domain, the DC supply V_{DD} is considered as an AC ground. Transistors M_1 and M_2 are replaced by their respective small-signal equivalents, including their output resistances r_{O1} and r_{O2} to provide a realistic estimation of the circuit's behavior. Therefore, the small signal circuit representation of Fig. 1, is shown in Fig. 2

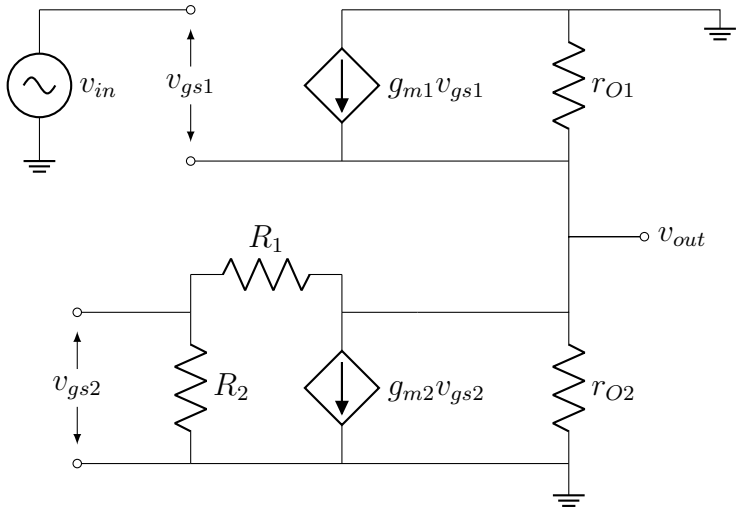


Figure 2: Complete small-signal equivalent circuit configuration featuring finite output resistances and local negative feedback network.

Given the circuit shown in Fig. 2, we can start doing the analysis of the small signal in order to calculate the voltage gain of it. We can do this process step by step in the following subsections.

2.1 Defining the Control Voltages

First, we must define the gate-to-source voltages (v_{gs}) that control the dependent current sources of transistors M_1 and M_2. For the input transistor M_1, the input voltage V_{in} is applied at the gate, and the output v_{out} is located at the source. Thus, the control voltage is:

$$v_{gs1} = v_{in} - v_{out} \tag{1}$$

For the active load transistor M_2, the gate voltage is defined by the voltage divider formed by R_1 and R_2, which is fed by the output node v_{out}. Therefore:

$$v_{gs2} = v_{out}\left(\frac{R_2}{R_1 + R_2}\right) \tag{2}$$

2.2 Applying Kirchhoff's Current Law

We apply Kirchhoff's Current Law at the v_{out} node, stating that the sum of the currents entering the node equals the sum of the currents leaving the node towards the AC ground. Therefore, the current entering the node from the dependent source of M_1 is:

$$i_{in} = g_{m1}v_{gs1} = g_{m1}(v_{in} - v_{out}) \tag{3}$$

The currents leaving the node are distributed through four parallel paths: the output resistance of M_1 (r_{O1}), the output resistance of M_2 (r_{O2}), the resistive feedback branch

$(R_1 + R_2)$, and the dependent current source of M_2. The total leaving current is:

$$i_{out} = \frac{v_{out}}{r_{O1}} + \frac{v_{out}}{r_{O2}} + \frac{v_{out}}{R_1 + R_2} + g_{m2} v_{gs2} \tag{4}$$

2.3 Analytical Expression for the Voltage Gain

Due to the fact that the entering and leaving currents have to be the same, this is the mathematical equivalent to equating Eq. 3 and Eq. 4 and thus:

$$g_{m1}(v_{in} - v_{out}) = \frac{v_{out}}{r_{O1}} + \frac{v_{out}}{r_{O2}} + \frac{v_{out}}{R_1 + R_2} + g_{m2} v_{gs2} \tag{5}$$

Substituting the expression for v_{gs2} found in Eq. 2, we have:

$$g_{m1}(v_{in} - v_{out}) = \frac{v_{out}}{r_{O1}} + \frac{v_{out}}{r_{O2}} + \frac{v_{out}}{R_1 + R_2} + g_{m2} \left(v_{out} \frac{R_2}{R_1 + R_2} \right) \tag{6}$$

By distributing the g_{m1} term in the left side of the Eq. 6 and grouping all the v_{out} terms on the right side of the equation, we obtain:

$$g_{m1} v_{in} = v_{out} \left(g_{m1} + \frac{1}{r_{O1}} + \frac{1}{r_{O2}} + \frac{1}{R_1 + R_2} + g_{m2} \frac{R_2}{R_1 + R_2} \right) \tag{7}$$

Finally, solving for the voltage gain $A_v = v_{out}/v_{in}$ yields the analytical expression:

$$A_v = \frac{g_{m1}}{g_{m1} + \frac{1}{r_{O1}} + \frac{1}{r_{O2}} + \frac{1}{R_1 + R_2} + g_{m2} \frac{R_2}{R_1 + R_2}} \tag{8}$$

2.4 Numerical Calculation

Using the given design parameters and the assumed output resistances ($r_{O1} = r_{O2} = 5\ [k\Omega]$), we can compute the final value of the voltage gain obtained in Eq. 8:

$$A_v = \frac{1\ [mS]}{1\ [mS] + \frac{1}{5\ [k\Omega]} + \frac{1}{5\ [k\Omega]} + \frac{1}{10\ [k\Omega] + 5\ [k\Omega]} + 3\ [mS] \frac{5\ [k\Omega]}{10\ [k\Omega] + 5\ [k\Omega]}} \approx 0.4054 \left[\frac{V}{V} \right] \tag{9}$$

3 Conclusion

The small-signal analysis yields a numerical voltage gain of approximately 0.4 $\left[\frac{V}{V}\right]$. This sub-unity gain is theoretically consistent with a *Source Follower* topology [1, p. 332]. The result demonstrates that the inclusion of non-idealistic (finite) output impedances (r_{O1} and r_{O2}), along with the active participation of M_2 through the local negative feedback network, significantly loads the output node, attenuating the ideal unity gain down to approximately 40.5 [%].

4 Bibliography

[1] B. Razavi, *Fundamentals of Microelectronics*, 2nd. John Wiley & Sons, 2013, ISBN: 978-1-118-15632-2.